

Annotated Formula Supplement

A Color-Coded Decoder for

Audit the Scaffold, Not the Checkpoint

How to read this supplement

This companion document reproduces every defining and result formula from the main paper and colors each term by its **semantic role**. A color means the same kind of thing *everywhere*: the confidence weight α_t is purple in the protocol, in the training-error proof, and in the algorithm alike. Structural operators ($\sum$, $\prod$, $\ln$, $\exp$) and plain indices (t, T, N, i, K) are left black to reduce clutter. Under each formula a short legend names its colored terms in plain English and points back to where the formula appears in the main text.

Color	Role	Typical symbols
■	Specifications / inputs / spec-distribution	$\mathcal{X}, x_i, \mathcal{D}, D_t, w_i$
■	Artifacts: drafts and optimal targets	$\mathcal{Y}, y_i, y, z_t$
■	Weak agent / hypothesis / operators	$h_t, H, \mathcal{H}, R_t, F_T$
■	Confidence weight / margin / normalizer	$\alpha_t, f(x), Z_t, J$
■	Error / edge / loss / thresholds	$\epsilon_t, \gamma_t, L, \theta, \delta$
■	Quality / value / improvement / ceiling	$V, V^*(W), \eta_t, q_{i,t}, B$
■	Empirical fit parameters (Part C)	$\rho, \sigma, \psi, \pi_{\text{pure}}$

Two glyph warnings. (1) Two quantities that a single τ would conflate are kept distinct: $\tau = 0.6$ is Part B's *pass threshold* (an error/edge quantity, rose), while Part C's *mean survivor-ratio statistics* are $\psi_{\text{Opus}}, \psi_{\text{Sonnet}}$ (empirical, olive). Part C's ρ is likewise the churn-decay factor only; its rank correlations are written r_s . (2) The success indicator flips: the theory writes $L(y, h(x)) > 0$ for *failure*, while the algorithm's weight update writes $L(\dots) = 0$ for *success* inside the exponent. Both encode the same AdaBoost update, read oppositely.

1 Problem Setup

$$L(y, \hat{y}), \quad f : \mathcal{X} \rightarrow \mathcal{Y}, \quad \mathcal{D} \text{ over } \mathcal{X} \times \mathcal{Y}.$$

$L(y, \hat{y})$

structured loss between a candidate artifact $\hat{y}$ and the optimal target y (e.g. token or AST edit distance).

$\mathcal{X}$

space of specifications (prompts / issue tickets).

$\mathcal{Y}$

space of optimal code artifacts (the targets).

$\mathcal{D}$

joint distribution over specification–solution pairs.

f

the structured mapping specifications $\rightarrow$ code that the system learns.

Main text: Problem Setup, § Theoretical Analysis.

2 Weak Coding Agent and Edge

$$\text{err}_{\mathcal{D}}(h_t) = \Pr_{(x,y) \sim \mathcal{D}} [L(y, h_t(x)) > 0] \leq \frac{1}{2} - \gamma_t.$$

h_t

the weak coding agent at round t : proposes a patch to improve the current draft.

$\text{err}_{\mathcal{D}}(h_t)$

its probability of failing (nonzero loss) on a spec drawn from $\mathcal{D}$.

$L(\dots) > 0$

the boolean failure criterion (loss > 0 = fail; $= 0$ = success).

γ_t

the *edge* — how much better than chance ($\frac{1}{2}$) the agent is. In code space even a small positive edge presupposes real capability.

Main text: Definition 1 (Weak Coding Agent); edge revisited in Remark 14.

3 Strong Coding System (Two Levels)

$$F_T(x) = (R_T \circ R_{T-1} \circ \dots \circ R_1)(F_0(x)) \quad (\text{artifact level})$$

$$f(x) = \sum_{t=1}^T \alpha_t h_t(x), \quad \alpha_t \in \mathbb{R}_{\geq 0} \quad (\text{score level: additive margin})$$

F_T

the artifact after T rounds — a *sequential composition* of patches (order-dependent, non-linear).

R_t

refinement operator applying patch h_t to the current draft.

F_0

the initial draft before any refinement.

$f(x)$

the additive confidence *margin* — a weighted vote over specifications; this is what inherits the AdaBoost machinery.

α_t

confidence weight assigned to agent t .

Main text: Definition 8 (Strong Coding System). Artifact vs. margin distinction (§ Framework).

4 Agentic Boosting Protocol

$$\epsilon_t = \Pr_{(x,y) \sim D_t} [L(y, h_t(x)) > 0], \quad \alpha_t = \frac{1}{2} \ln \frac{1 - \epsilon_t}{\epsilon_t},$$

$$D_{t+1}(i) = \frac{D_t(i) \exp(-\alpha_t y_i h_t(x_i))}{Z_t}.$$

ϵ_t weighted error of agent t under the current spec-distribution D_t .

α_t confidence weight — larger when the error is smaller (the AdaBoost rule).

$D_t(i)$ weight on spec i at round t ; hard specs get up-weighted.

Z_t normalizer keeping D_{t+1} a probability distribution.

Main text: Definition 9 (Agentic Boosting Protocol).

5 Training-Error Bound

$$\frac{1}{N} \sum_{i=1}^N \mathbf{1}[H(x_i) \neq y_i] \leq \prod_{t=1}^T Z_t, \quad Z_t = 2\sqrt{\epsilon_t(1 - \epsilon_t)}.$$

H the strong system's decision (sign of the margin f).

$\mathbf{1}[H \neq y_i]$ training mistake indicator for spec i .

$\prod_t Z_t$ product of per-round normalizers — the bound on training error.

Z_t shrinks below 1 whenever the agent has positive edge, so the product decays.

Main text: Theorem 10 (Training Error of Agentic Boosting).

6 Exponential Decay of Training Error

$$\text{err}_{\text{train}}(H) \leq \exp(-2\gamma^2 T).$$

$\text{err}_{\text{train}}(H)$ fraction of training specs the strong system gets wrong.

γ uniform edge lower bound ($\gamma_t \geq \gamma > 0$ for all t).

T number of rounds; error falls exponentially in T once the edge holds.

Main text: Corollary 11 (Exponential Decay). Applies only once agents cross the capability threshold (Remark 14).

7 Generalization Bound

$$\Pr_{(x,y) \sim \mathcal{D}} [H(x) \neq y] \leq \Pr_{\text{train}} [y f(x) \leq \theta] + O\left(\sqrt{\frac{d \ln(N/d) \ln(1/\theta) + \ln(1/\delta)}{N}}\right).$$

$\Pr_{\mathcal{D}}[\dots]$

true (test) error over the data distribution.

$y f(x) \leq \theta$

training margin-violation term; shrinks as boosting grows margins.

θ

margin threshold (> 0).

d

VC-dimension of the weak-agent class $\mathcal{H}$.

δ

failure probability (bound holds with probability $1 - \delta$).

N

number of i.i.d. training samples. The bound is *independent of* T .

Main text: Theorem 12 (Generalization Error). Margin bound of Schapire et al. (1998).

8 Equivalence to Forward Stagewise Additive Modeling

$$J(\alpha, h) = \sum_{i=1}^N \exp\left(-y_i(f_{t-1}(x_i) + \alpha h(x_i))\right), \quad L(y, f(x)) = e^{-y f(x)}.$$

$J(\alpha, h)$

exponential-loss objective minimized greedily each round.

f_{t-1}

margin accumulated through round $t - 1$.

αh

the new term added this round (weight $\times$ agent).

$L = e^{-y f}$

the exponential loss whose greedy coordinate descent *is* the protocol.

Main text: Theorem 13 (Boosting as Greedy Coordinate Descent).

9 The Refinement Game

$$\mathbb{E}_{y \sim \mathcal{D}(\cdot|x)} [V(R_t(z, a_t^*))] \geq V(z) + \eta_t \quad (\text{improvement property})$$

$$\mathbb{E}[V(z_T)] \geq V(z_0) + \sum_{t=1}^T \eta_t, \quad \sum_{t=1}^{\infty} \eta_t \leq 1 - V(z_0) \quad (\text{saturation}).$$

V

quality function $\mathcal{Y} \rightarrow [0, 1]$ — how good the current artifact is. Note the domain: quality is scored on *artifacts*, not on the specification space $\mathcal{X}$.

$R_t(z, a_t^*)$

best refinement of draft z under action a_t^* (patch / refactor / test).

η_t

guaranteed minimum quality gain at round t (analog of the edge γ_t).

z_0, z_T

initial and round- T drafts, both artifacts in $\mathcal{Y}$.

$$\sum \eta_t \leq 1 - V(z_0)$$

total improvement is capped: gains must vanish, $\eta_t \rightarrow 0$.

Main text: Definition 1 (Weak Coding Agent), which introduces R_t , and Theorem 2 (Monotone Improvement under Refinement).

10 High-Probability Refinement

$$\Pr \left[V(z_T) \geq V(z_0) + \sum_{t=1}^T \eta_t \right] \geq 1 - \sum_{t=1}^T p_t.$$

p_t per-round probability that the agent *fails* to deliver the gain η_t (a quality regression).

$$1 - \sum p_t$$

union-bound guarantee that all rounds improve, relaxing the strict expectation.

Main text: Corollary 15 (High-Probability Refinement). Verified against Mathlib’s probability library.

11 Stationarity Dichotomy

(i) fixed ceiling $V(z) \leq B$: $\sum_{t=1}^{\infty} \eta_t \leq B - \mathbb{E}[V(z_0)] \Rightarrow \eta_t \rightarrow 0$ (saturation).

(ii) sustained edge $\eta_t \geq c > 0$: $\forall B \in \mathbb{R}, \exists T$ s.t. $\mathbb{E}[V(z_T)] > B$ (divergence).

B a fixed upper bound (ceiling) on quality.

c a uniform positive per-round gain the expanding class must sustain to diverge.

(i) *stationary* regime — fixed reachable class $\Rightarrow$ diminishing returns.

(ii) *non-stationary* regime — only a perpetually expanding class escapes the plateau.

Main text: Proposition 3 (Stationarity Dichotomy). Formally verified in Lean (`Proposition8.lean`).

12 The Frozen-Weight Ceiling

(M1) $\mathcal{H}_t(C)$ expandable by scaffold rewrites; (M2) $V(z) \leq V^*(W)$ for all reachable z .

$$\sum_{t=1}^{\infty} \eta_t \leq V^*(W) - \mathbb{E}[V(z_0)], \quad \mathbb{E}[V(z_t)] \nearrow L \leq V^*(W).$$

$\mathcal{H}_t(C)$

effective hypothesis class of reachable patches, a function of the mutable scaffold C .

W the model weights (frozen).

$V^*(W)$

the ultimate quality ceiling achievable with weights W by *any* scaffold.

L the limit the system saturates to — no greater than the ceiling.

Main text: Corollary 4 (Frozen-Weight Ceiling): frozen weights guarantee the ceiling $V^*(W)$, not a safe trajectory to it. Lean: `cor_frozen_weight_ceiling`.

13 Orchestration Selects a Ceiling; It Does Not Raise One

$$V_{\text{orch}}^* = \max_i V^*(W, C_i) \quad (\text{exact equality, unconditional}).$$

$$\mathbb{E}[V(z_T)] \geq V(z_0) + \sum_{t=1}^T \max_i \eta_t^{(i)}, \quad \sum_{t=1}^T \max_i \eta_t^{(i)} \leq 1 - V(z_0).$$

C_i worker i 's scaffold — distinct prompts, tools, or specializations, possibly sharing weights W .

$V^*(W, C_i)$ worker i 's tight ceiling, $\sup\{V(z) : z \text{ reachable under } C_i\}$.

V_{orch}^* what best-of- k selection attains. The $=$ is the content: orchestration *realizes* the best worker's ceiling and never exceeds it, even when no single worker's reachable class contains the union.

$\max_i \eta_t^{(i)}$ the orchestrator banks each round's best gain, dominating every worker's depth-only guarantee...

$\leq 1 - V(z_0)$...yet the total is still capped by the same bounded metric: **width buys rate, not budget**.

Main text: Proposition 5 (Orchestration Selects a Ceiling). The equality is where the formalization earned its place: a conjectured *strict* inequality was refuted by two disjoint singleton classes of equal value. Width half stated separately as Prop. 18, whose extra hypothesis (monotonicity of the inner action-expectation) the formalization flags.

14 Bounded Overlap Characterizes a Correct Vote

$$m_i := \#\{0 \leq t < k : y_i h_t(x_i) = -1\}, \quad m^* := \max_i m_i.$$

$$y_i f(x_i) = k - 2m_i \quad \implies \quad \text{vote error} = 0 \iff 2m_i < k \quad \forall i \iff m^* < k/2.$$

measured (Part B): $m^* = k$ at every round.

m_i how many of the k workers fail point i . The index runs $0 \leq t < k$, matching the formalization's range k ; it counts all k workers, not $k - 1$.

m^* the *tight overlap* — one integer that decides whether a vote can work at all.

$y_i f(x_i) = k - 2m_i$ splitting the unweighted margin at point i ; this identity is the whole proof.

$m^* < k/2$ *exact*, not merely sufficient — and sharp: one point at $2m_i=k$ already makes the error nonzero, and if every point sits there the error is 1.

$m^* = k$

our pool at the ceiling of overlap: some task defeats *every* worker, so the red mass is the vote's error.

Main text: Proposition 6 (Bounded Overlap Characterizes a Correct Vote), measured in § Theoretical Analysis (Fig. 1). Weighted voting is the one boosting-*exact* arrangement (Prop. 17), which is why this condition, not the ceiling, is what binds a verifier-free orchestrator. Lean: `vote_margin_eq`, `thm_overlap_iff_zero_error`, `cor_tight_overlap_iff`, `thm_overlap_boundary_sharp`.

15 The Reused-Pool Dichotomy

$\epsilon_0 := \min \left\{ \text{zero-one training error of } \sum_j A_j H_j : A_j \geq 0 \right\}$ (a property of the pool alone).

$$(i) \quad \epsilon_0 > 0 : \quad \epsilon_0 \leq e^{-2\gamma^2 T} \quad \implies \quad T \leq \frac{\log(1/\epsilon_0)}{2\gamma^2}.$$

$$(ii) \quad \epsilon_0 = 0 : \quad \forall D \exists j : \quad \sum_{i : y_i H_j(x_i) = -1} D(i) \leq \frac{1}{2} - \gamma, \quad \gamma = \frac{\theta}{2S} > 0.$$

ϵ_0 best zero-one training error any *nonnegative combination* of the pool attains — the quantity that decides which branch applies.

(i) the *horizon*: the edge hypothesis provably fails by a finite, computable round, uniformly over the schedule and the confidences, depending on k only through ϵ_0 .

θ, S interpolation margin $\min_i y_i \sum_j A_j H_j(x_i)$ and coefficient sum $\sum_j A_j$ of the witnessing combination.

(ii) the edge $\gamma = \theta/2S$ is fixed *before* the distribution is chosen, so the hypothesis survives every round of any run.

exhaustive

the two cases are mutually exclusive and cover everything: **a fixed pool is not a renewable resource.**

Main text: Proposition 7 (Reused-Pool Dichotomy); halves stated with their exact hypotheses as Props. 20 and 21. Only the interpolating direction is formalized — the converse needs AdaBoost's minimax duality and remains unformalized.

16 Framework Mapping (Boosting $\leftrightarrow$ Coding)

$$\underbrace{X}_{\text{input}} \mapsto \underbrace{Y}_{\text{target}}, \quad \underbrace{Y - F_{t-1}(X)}_{\text{residual}} \approx h_t, \quad \text{loss } L(Y, \hat{Y}), \quad \text{distribution } D_t.$$

$X \mapsto Y$

structured-prediction view: a specification maps to its optimal code.

$F_{t-1}(X)$

current draft after $t - 1$ rounds.

$Y - F_{t-1}(X)$

the *residual error* each weak agent h_t predicts (a git diff / patch).

Main text: Table 1 (Framework mapping), § Framework.

17 Algorithm: Agentic Boosting Protocol for Specifications

Require: $\{(x_i, y_i)\}_{i=1}^N$, T spec-code pairs, rounds
 $w_i \leftarrow 1/N$ initialize weights
 $\epsilon_t = \sum_i w_i \mathbf{1}[L(y_i, h_t(x_i)) > 0]$ select h_t minimizing weighted error
 if $\epsilon_t \geq \frac{1}{2}$: stop no weak learner available
 $\alpha_t \leftarrow \frac{1}{2} \ln \frac{1 - \epsilon_t}{\epsilon_t}$ confidence weight
 $w_i \leftarrow w_i \cdot \exp(-\alpha_t \mathbf{1}[L(y_i, h_t(x_i)) = 0])$ re-weight (note: $=0$ = success)
 $w_i \leftarrow w_i / \sum_j w_j$ normalize
return $H(x) = (R_T \circ \dots \circ R_1)(z_0)$, $f(x) = \sum_t \alpha_t h_t(x)$ artifact + margin

w_i unnormalized weight on spec i (the operational form of D_t).
 $\mathbf{1}[L = 0]$ success indicator — note this flips the theory’s > 0 (failure) convention.
 return artifact H by sequential composition; confidence by additive margin f .

Main text: Algorithm 1 (Appendix). Operational form of Definition 9.

18 Part B: SWE-bench Regime Estimators

$$q_{i,t} \in [0, 1], \quad \tau = 0.6, \quad \text{err}_t = \frac{1}{K} \sum_i \mathbf{1}[\max_{s \leq t} q_{i,s} < \tau] \approx \exp(-2 \sum_s \gamma_s^2).$$

$$S_{i,t} = \sum_{s \leq t} \max(0, q_{i,s} - q_{i,s-1}) \leq 1 - q_{i,0}, \quad m_{i,t} = 2q_{i,t} - 1.$$

$q_{i,t}$ binary harness pass/fail of task i at round t (1 = all FAIL_TO_PASS pass, no regressions).
 $\tau = 0.6$ pass *threshold* on the $[0, 1]$ partial score $q_{i,t}$ — the only meaning τ now carries.
 err_t running-best pass error over K tasks; tracks the exponential-decay prediction.
 $S_{i,t}$ cumulative improvement, bounded by headroom $1 - q_{i,0}$ (refinement saturation).
 $m_{i,t}$ margin per task; predicted to shift rightward across rounds.

Main text: §Part B (Quality signal; Predictions), $K=55$, $T=4$.

19 Part C: Production-Session Empirical Model

$$\text{survivor ratio} = \frac{|\text{net lines}|}{\text{total churn}}, \quad \psi_{\text{Opus}} \approx 0.55 \text{ vs. } \psi_{\text{Sonnet}} \approx 0.55.$$

$$\log(\text{churn}_{t+1}) \sim \mathcal{N}(\rho \log(\text{churn}_t), \sigma^2), \quad \rho = 0.77, \sigma = 2.39, \quad \pi_{\text{pure}} \approx 0.071.$$

survivor ratio

structural proxy: net lines retained over total churn (`git_session_extractor.py:152`).

$\psi_{\text{Opus}}, \psi_{\text{Sonnet}}$

mean non-pure survivor ratios by model tier — the Part-C meaning of τ (a statistic, not a threshold). Gap = -0.003 (95% CI $[-0.075, +0.069]$), a null surviving four independent robustness checks — not a directional effect.

ρ

geometric churn-decay rate (log-normal AR(1) coefficient); < 1 signals convergence.

σ

spread of the log-normal churn model.

π_{pure}

mixture weight for pure-addition sessions (survivor ratio = 1).

Main text: §Part C, Findings 1–3. Ties ρ to Theorem 2 decay and ψ to the $V^*(W)$ ceiling (Corollary 4).